

Vinayak Gajjewar

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Education	<i>Ph.D.</i> , Computer Science University of California, Santa Barbara - GPA: 4.0/4.0 - Advisors: Chandra Krintz and Rich Wolski - Research topics: Internet of Things, distributed systems, geospatial computing - Selected coursework: Scalable Internet Services (291A), Runtime Systems (263), Blockchain and Distributed Systems (293B), Operating Systems (270), Program Analysis (260)	Expected June 2028
	<i>Bachelor of Science</i> , Computer Science University of California, Riverside	Graduated June 2023
Skills	C, C++, Java, Scala, JavaScript/TypeScript Unix/Linux, Slurm, Django, Docker+Kubernetes, Git, Spark, Hadoop, Node.js, MongoDB	
Publications	Gajjewar, Vinayak, Rich Wolski, and Chandra Krintz. "Ariadne: A Resilient IoT Storage Service for Geospatial Computing." IC2E 2026. 17% acceptance rate!	
	Gajjewar, Vinayak, Rich Wolski, and Chandra Krintz. "RIoTstore: Resilient Data Storage for Spatial IoT Applications." 2025 IEEE Annual Congress on Artificial Intelligence of Things (AIoT). IEEE, 2025.	
	Singla, Samridhi, Ayan Mukhopadhyay, Michael Wilbur, Tina Diao, Vinayak Gajjewar, Ahmed Eldawy, Mykel Kochenderfer, Ross Shachter, and Abhishek Dubey. "Wildfiredb: An open-source dataset connecting wildfire occurrence with relevant determinants." In NeurIPS Thirty-fifth Annual Conference on Neural Information Processing Systems. 2021.	
Experience	<i>Graduate Student Researcher</i> RACELab, UC Santa Barbara, CA - Building fault-tolerant distributed data storage layer for remote IoT environments - Improving energy efficiency for high-availability geospatial computing applications - Achieved 99.99% SLA on failure-prone deployment of >200 Raspberry Pis	September 2023 – present
	<i>Software Products Intern</i> Esri Inc., Redlands, CA Used TypeScript, CloudFormation, and Electron to develop an application for deploying ArcGIS Enterprise to AWS.	Summer 2024
	<i>Undergraduate Researcher</i> UC Riverside Big-Data Lab, Riverside, CA Contributed to the development of Raptor, a Raster + Vector query processing engine written in Java and Spark for manipulating and visualizing geospatial data.	Summer 2023
	<i>Software Products Intern</i> Esri Inc., Redlands, CA	Summer 2022

	• Used Node.js to write a framework for connecting remote data sources (PostgreSQL, REST APIs) to Esri software ecosystem. • Wrote technical documentation and code samples for new features of ArcGIS Enterprise.	
	<i>Research Fellow</i> UC Riverside Big-Data Lab, Riverside, CA	May 2021 - May 2022
	• Used Maven and Apache Spark to build a scalable analytics system that uses satellite data to compute wildfire spread, resulting in a 2 order of magnitude performance increase over the state of the art.	
	<i>Instructor</i> iD Tech Camps, Santa Clara, CA	Summer 2020
	• Tutored 50+ K-5 children on various topics in computer science, from basic programming concepts to video game development.	
	<i>Computer Science Intern</i> SchoolCity Inc., Santa Clara, CA	Summer 2018
	• Developed a data analytics application that extrapolates patterns in School City product usage across school districts using MongoDB and Express.js.	
Teaching experience	<i>UCSB CMPSC 170: Operating Systems</i> Teaching Assistant	Winter 2024, Spring 2024
	• Worked with Profs. Rich Wolski (W24) and Tao Yang (S24) • Subjects: processes, inter-process communication, I/O, file systems, memory management • Debugged OS implementations, held office hours and remedial sessions,	
	<i>UCSB CMPSC 190B: IoT Systems</i> Teaching Assistant	Fall 2023
	• Worked with Prof. Chandra Krintz • Subjects: IoT fundamentals, software architectures, communication protocols, security concerns, distributed + multi-tier (sensors+edge+cloud) programming • Managed hardware (Raspberry Pi + Arduino), debugged implementations, held office hours and remedial sessions	
Presentations	2025 — Spatial awareness for scalable IoT systems. (2025). Major Area Exam. Santa Barbara, CA. 2023 — Implementing a Distributed Evapotranspiration Model. (2023). 2023 UCR Undergraduate Research & Creative Activities Symposium. Riverside, CA. 2021 — Increasing the Efficiency of Geospatial Data Processing. (2021). Research in Science & Engineering Symposium. Riverside, CA.	
Awards & honors	2023 — ASA, CSSA, SSSA Outstanding Senior (19 seniors recognized nationally) 2021 — Digital Agriculture Fellowship (Artificial Intelligence for Sustainable Agriculture)	
Selected press	Ober, Holly. (December 8, 2021.) Wildfire dataset could help firefighters save lives and property. UCR News Archive. https://news.ucr.edu/articles/2021/12/08/wildfire-dataset-could-help-firefighters-save-lives-and-property	